

WEEK 1 PROJECT REPORT

Ludo Star Game Using Java

Student Information

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Project Title: Ludo Star Game

Course: Software Development I

Week 1

1. Introduction

Ludo Star is a digital version of the traditional Ludo board game. The game is played by 2 to 4 players, where each player rolls a dice and moves their tokens around the board. The objective is to move all tokens from the starting area to the home area before the opponents.

This project aims to develop a desktop-based Ludo Star game using Java. The project will help in understanding Object-Oriented Programming (OOP), game development concepts, and graphical user interface design.

2. Project Objectives

The main objectives of this project are:

- Develop a Ludo Star game using Java.
 - Implement standard Ludo game rules.
 - Create a user-friendly interface.
 - Support multiple players.
 - Implement dice rolling and token movement.
 - Apply Object-Oriented Programming concepts.
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3. Requirement Analysis

Functional Requirements

The system should:

- Start a new game.
- Allow 2–4 players.
- Roll a dice randomly.
- Move tokens according to dice values.
- Manage player turns.
- Detect the winner correctly.

Non-Functional Requirements

- Easy to use interface.
- Fast response time.
- Reliable game logic.
- Organized and maintainable code.

4. Technology Stack

Component	Technology
Programming Language	Java
IDE	Visual Studio Code
GUI Framework	Java Swing (Future Implementation)
Platform	Windows

5. System Design

The project is divided into the following classes:

Main Class

Starts and controls the execution of the program.

Board Class

Represents the game board.

Player Class

Stores player information such as name and color.

Token Class

Represents the player's token and its position.

Dice Class

Generates random dice values.

GameController Class

Controls the overall game flow.

6. Week 1 Work Completed

During Week 1, the following tasks were completed:

- Studied the rules and gameplay of Ludo Star.
- Analyzed project requirements.
- Selected Java as the development language.
- Created the project structure.
- Designed the initial class architecture.
- Implemented basic Java classes:
 - Main.java
 - Board.java
 - Player.java
 - Token.java
 - Dice.java
 - GameController.java
- Connected all classes through the Main class.
- Successfully executed the project setup.

7. Coding Progress

The basic project structure has been created and all classes have been connected successfully.

Current implemented features:

- Board initialization
- Player creation
- Token initialization
- Dice rolling
- Game controller initialization

Sample Output:

```
=====
LUDO STAR GAME USING JAVA
=====
```

Ludo Board Initialized Successfully.

Players Created:

Player 1 - Red

Player 2 - Blue

Tokens Initialized:

Token 1 Position: 0

Token 2 Position: 0

Dice Rolled: 4

Game Controller Started.

Week 1 Project Setup Complete.

8. Challenges Faced

- Understanding Java project structure.
 - Learning how different classes interact with each other.
 - Setting up the development environment in Visual Studio Code.
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10. Conclusion

Week 1 focused on project planning, requirement analysis, and setting up the development environment. The basic structure of the Ludo Star game was successfully created using Java. All necessary classes were implemented and connected, providing a solid foundation for future development. The next phase of the project will focus on graphical user interface development and board design.